

are curious about intelligence, but we do not know much about our brains and where does it actually come from.

People have become more curious about the intelligence itself through the years of exploration, which has led them to build intelligent systems we today call artificial intelligence (AI). The history of AI can be traced back to a summer workshop back in 1956. It is widely considered as the founding event of AI as a field.

According to Wikipedia, "It is a theory and development of computer systems able to perform tasks normally requiring human intelligence, such as visual perception, speech recognition, decision-making, and translation between languages."

Evolution of computing power

The chart in Fig. 2 is from Dr Raymond Kurzweil's book that was published back in 1990, titled "The age of intelligent machines." This chart summarises the computing power measured as millions of instructions per second. The dots in the chart represent a curve.

This curve suggested that by the year 2020 the computing power could reach the intelligence level of monkeys and human beings. May be, right now this claim is too bold to go with. But the bottom line is that a sufficiently advanced computer program could exhibit human-level intelligence.

Major limitations

Some limitations have emerged from the intrinsic architecture of the computing system hardware. In the past thirty years, people building computing systems have had different objectives for improving the RAM and processors. They aimed to improve the execution speed for processors while the major goal for the memory part was to increase the storage density. The two components have a huge performance gap to this day.

Another important aspect is the limited power efficiency ceiling. The power efficiency ceiling is much

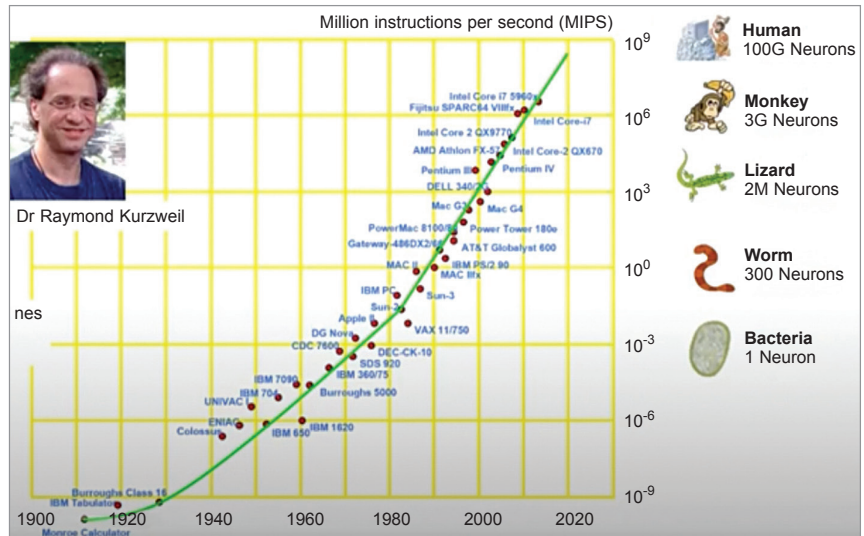


Fig. 2: Evolution of computing power

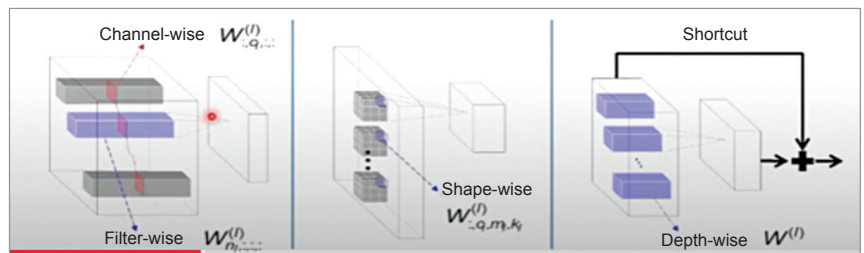


Fig. 3: Structurally sparse deep neural network

Step 1: Weights are split to G groups $\mathbf{w}^{(1..G)}$

e.g. $(w_1, w_2, w_3, w_4, w_5) \rightarrow$ group 1: (w_1, w_2, w_3) , group 2: (w_4, w_5)

Step 2: Add group Lasso on each group $\mathbf{w}^{(g)}$ $\|\mathbf{w}^{(g)}\|_g = \sqrt{\sum_{i=1}^{|w^{(g)}|} (w_i^{(g)})^2}$ i.e. vector length $\sqrt{w_1^2 + w_2^2 + w_3^2}$, $\sqrt{w_4^2 + w_5^2}$

Step 3: Sum group Lasso over all groups as a regularization: $R_g(\mathbf{w}) = \sum_{g=1}^G \|\mathbf{w}^{(g)}\|_g$, $\sqrt{w_1^2 + w_2^2 + w_3^2} + \sqrt{w_4^2 + w_5^2}$

Step 4: SGD optimizing $\arg \min_{\mathbf{w}} \{E(\mathbf{w})\} = \arg \min_{\mathbf{w}} \{E_D(\mathbf{w}) + \lambda_g \cdot R_g(\mathbf{w})\}$

Fig. 4: Group lasso regulation

lower than the efficiency provided by human brains. Human brain is indeed the most efficient computing machine there is. So, let us take a quick look at it.

The human brain, on average, contains 15 to 30 billion neurons. It essentially has very complex structures and offers complicated operations. But its power consumption is extremely low at around 20 to 35 watts.

Fig. 3 shows that a human brain contains some gray matter for thinking and white matter for signal transi-

tions. The gray matter consists of a neocortex, which contains six layers where signals travel within the layers. The fundamental elements of the neural cortex are neurons and synapses.

There are three characteristics that can be found in human brain but not in the computers:

Low power consumption. Only about 35 watts of power is consumed by the brain versus roughly 250 watts by a graphics processing unit (GPU) for rigid operations.

Fault tolerance. The cells within